



# Ryan Zernach Senior AI Systems Engineer • Agentic Infrastructure & Product Engineering

Portfolio: [Ryan.Zernach.com](https://ryan.zernach.com) • Email: [Ryan@Zernach.com](mailto:Ryan@Zernach.com) • [LinkedIn](#) • [GitHub](#) • [Remote](#)

## HIGHLIGHTS

- Distributed systems engineer with 10+ years of experience in Agents, AI Infrastructure, Mobile Apps, Back-End Systems, & MLOps
- Scaled CellarChat™ tool-calling agent to 1M+ active members, complete with RLHF & CI/CD LLM evaluation scoring pipelines
- Engineered Apple Pay subscriptions to lift recurring revenue by 25%+ which lifted mobile app ratings to 4.9★ (10k+ reviews)
- Completed Gauntlet AI engineering fellowship with 2% acceptance rate & mastered the art of agentic AI orchestration
- Designed observable backend systems that support autonomous AI-powered VoIP eCommerce payment automations
- Awarded Top 10 of 10,000 international projects in 2024 NASA Space Apps hackathon with AI Voice Assistant

## TECH STACK

### Programming Languages

TypeScript, Python, Rust, Golang, Java, C#, Ruby, Swift, Kotlin, Bash, JavaScript, PHP, C++, Objective-C, R, Solidity, Dart, Flutter, XML

### AI Systems & Machine Learning

LLMOps, RAG, Agents, MCP, Evaluations, Semantic Kernel, LangChain, Langfuse, ML Pipelines, Computer Vision, LiteLLM, Vector Embeddings, PyTorch, Knowledge Graphs, TensorFlow, Scikit-Learn, Bedrock, Foundry

### Platform & Backend Engineering

DevEx, Node.js, FastAPI, Django, Nest.js, Express, .NET Core, OpenAPI, Microservices, REST, GraphQL, WebSockets, Webhooks, gRPC, Kafka

### Data & Persistence

PostgreSQL, SQL, NoSQL, Redis, Vector Databases, ETL, Snowflake, Graph Databases, Entity Relationships, Caching, BigQuery, PySpark

### Cloud Infrastructure & DevOps

AWS, Azure, GCP, Cloudflare, Docker, GitHub Actions, Terraform, Kubernetes, CI/CD, CloudOps, Blue/Green, DNS, Networking

### Front-End & Product Engineering

iOS, Android, Web, Swift, Kotlin, React, React Native, Flutter, Expo Router, Next.js, Redux, Streamlit, Animations, Design, User Feedback

### Distributed Systems & Operations

Queues, Redis, RabbitMQ, Analytics, Pub/Sub, Linux, Airflow, Event-Driven Architecture, VoIP, Observability, Admin Systems, WebRTC, Rate Limiting, Horizontal Scaling, Load Balancing, Cron

### Security, Identity & Compliance

AuthN, AuthZ, OAuth, SOC 2, Identity, Role-Based Access Control, JWT, Secure Systems Design, SSO, API Security, Secrets Management

### Payments, Communications & Integrations

Stripe, Apple Pay, IAP, Billing, Subscriptions, Wallets, Twilio, SMS, Push Notifications, HTTP APIs, Marketing Tech, Email SMTP/IMAP

### Blockchain & Web3

Ethereum, Solana, dApps, WalletConnect, MetaMask, Polygon, Solidity, Smart Contracts, Ethers.js, Tokens, ERC-20, ERC-721

## EXPERIENCE

### Fractional AI Systems Engineer • Freelance • United States • Remote

July 2025 - Present

- Building AI-enabled products and back-end systems across fintech, semantic intelligence, and now focused on healthcare
- Dockerized semantic intelligence layer around MCP, OpenClaw, Plaid financial data, OSS, and federal tax preparations
- Published TranslateWallpaper, a cross-platform mobile app powered by LLM-generated language-learning wallpapers

### Senior AI Systems Engineer • CellarTracker • Seattle • Remote

Aug 2023 - Jul 2025

- Scaled agentic CellarChat™ to 1M+ international members & released world's highest-rated #1 wine app
- Architected OpenAPI pipeline, embeddings, fine-tuning, LLM evaluations, RLHF, and admin observability
- Integrated intelligent ML-powered recommendations for world's largest online database of wine reviews
- Maintained Python, TypeScript, React Native, Redis, C#/.NET, Microsoft Azure AI Foundry, CI/CD pipelines
- Helped to boost app store ratings to 4.9★ average with 10,000+ verified 5-star ratings on iOS and Android
- Embedded Apple Pay subscriptions in the React Native app to generate additional 25%+ recurring revenue
- Developed backend Python systems supporting agentic workflows, support tickets, & account management
- Improved user's experience of cellar management CRUD workflows to improve consistency and reduce taps
- Implemented animations, barcode scanning, activity feed posts, support ticket refactors, & OCR image search
- Streamlined the Redux store for fast, efficient, and persistent global state management between app sessions
- Migrated legacy support ticket system from Active Server Pages (ASP) to safer, type-friendly Express framework
- Rebuilt the React Native app's analytics implementations for tracking critical user interactions and in-app behavior
- Arranged A/B tests during paywall & onboarding experiences using phased rollout schedules for marketing analysis

**Back-End / Full-Stack Mobile Engineer** • Archlife Industries Software • Miami • Remote **Sept 2021 – Aug 2023**

- Delivered eTrading exchange for stocks/crypto, including public leaderboards and copy friend's portfolio distribution
- Used AWS Terraform to reliably deploy Java with Springboot, docker images, and admin observability dashboard
- Constructed core mobile app functionality, including: Plaid connections, trending investors, & suggested stocks
- Built social network features, such as direct chat, user profiles, posts, likes, emojis, & threaded comments
- Wrote the data-visualization logic for viewing and comparing the performance of different stocks/crypto
- Transitioned eSports React Native codebase from JavaScript to TypeScript for long-term maintainability
- Improved Ruby on Rails for push notifications, chat, & native calendar sync to improve user satisfaction
- Remedied bluetooth connection & map performance for Swift iOS/iPad telecommunications app

**Data Science & Machine Learning Tutor** • edX • Vanderbilt, UC Berkeley, +24 others • Remote **July 2020 – Sept 2021**

- During pandemic, taught 900+ online college students Python, back-ends, machine learning, TypeScript, websites, & databases
- Managed busy schedule of remotely teaching 30+ university students per week in 1:1 setting (one hour per person)
- Dozens of universities were subscribed to my employer's curriculum and online tutoring resources for students

**React Web Developer (Full-Stack)** • Georgia Hybrid Landscaping • Alpharetta, Georgia **Oct 2015 – Jan 2019**

- Engineered uploading videos/images, survey questions, pricing proposal, & processing payments via Java Spring Boot
- Migrated websites from PHP to TypeScript React, which successfully processed \$500k+ gross sales in three years
- Managed online eCommerce payments, visitor analytics, Google AdSpend, and email marketing campaigns
- Configured outbound marketing email campaign schedulers & listeners using IMAP/POP
- Setup observability dashboards for payments, marketing, analytics, API logs, & admin

**Payments Systems Intern (Card Processing)** • Cost Plus Processing • Alpharetta, Georgia **Sept 2012 – Jan 2013**

- Worked with merchant processing systems & payment gateways to understand critical financial flows
- Analyzed end-to-end card transaction lifecycles (interchange rates, risk) across payment networks
- Built foundational knowledge of payment infrastructure, interchange fees, & transaction routing

## EDUCATION & TRAINING

**AI Engineering Fellowship (G3)** • Gauntlet AI • Remote • Austin, TX **2025**

- Mastered how to command agents, automate workflows, and 100x enterprise-grade software engineering
- Harnessed the power of RAG, multi-agent systems, OpenAPI, and MCP to build agentic automations

**Data Science & Machine Learning Bootcamp** • Lambda School • Y Combinator • Remote **2020**

- Recommendation Engines, Computer Vision (CV) Models, Time-Series Forecasting, Natural-Language Processing
- Python, SciKit-Learn, PySpark, Django, Flask, FastAPI, Data Engineering, PostgreSQL Databases, Back-End APIs

**Turfgrass & Landscape Management** • University of Georgia (NALP) • Athens, Georgia **2015**

- Design Principles, Science of Grass, Outdoor Business Management, Landscape Architecture

**Computer Science & Mathematics Coursework** • Georgia Southern University • Statesboro, Georgia **2012**

- Visual Basic, Calculus, C, C++, R, Statistics, Data Structures, Programming Principles, Linear Regression

## CERTIFICATIONS

**2025** • Gauntlet AI Engineering Fellowship (G3)

**2024** • NASA Space Apps Hackathon Finalist

**2022** • Blockchain Hackathon Participant

**2020** • TensorFlow Computer Vision

**2020** • AutoML Pipelines (DataRobot Academy)

**2019** • Google Cloud Platform GCP Developer

**2019** • AWS Microservices Cloud Practitioner

**2019** • Y Combinator Startup School